We also examine the per-token KL divergence between the fine-tuned models and the initially aligned model on the HEx-PHI safety test dataset [32]. Specifically, for each harmful instruction $\tilde{\boldsymbol{x}}$, we take the outputs $\tilde{\boldsymbol{y}} \sim \pi_\theta(\,\cdot\,|\tilde{\boldsymbol{x}})$ from the fine-tuned model π_θ. Then, for each token position t, we compute the KL divergence $D_{\mathrm{KL}}\big(\pi_\theta(\,\cdot\,|\tilde{\boldsymbol{x}}, \tilde{\boldsymbol{y}}_{<t}) \,\big\|\, \pi_{\mathrm{aligned}}(\,\cdot\,|\tilde{\boldsymbol{x}}, \tilde{\boldsymbol{y}}_{<t})\big)$.

Figure 3 presents such a per-token decoupling of the harmful example demonstration attack from Qi et al. [22]. Here, a safety-aligned model (Llama-2-7B-Chat in our case) is fine-tuned on 100 (harmful instruction, harmful answer) data pairs, with a learning rate of 2×10^{-5} and a batch size of 64. Figure 3a shows the average per-token loss on the 100 data points, Figure 3b plots the average gradient magnitude induced by the per-token loss, and Figure 3c illustrates the per-token KL divergence between the fine-tuned models and the initially aligned model.

Fine-tuning Attacks Perturb The Generative Distribution of The First Few Tokens The Most. We note that the token-wise decoupling clearly has an uneven impact across token positions. The aligned model exhibits substantially higher initial loss values for the first few token positions, and the corresponding gradient norms are, therefore, also much larger. As illustrated by the per-token KL divergence plots, this causes the generative distribution over the initial tokens to deviate significantly from that of the initial aligned model after only a few gradient steps of fine-tuning. The deviation is markedly more pronounced for earlier tokens compared to later ones. Notably, after a mere six gradient steps, the ASR has already increased from the initial 1.5% to 87.9%. While we previously showed that the alignment of current models seems to largely be constrained to the first few tokens, this may also make it easy it unlearn safety behaviors during fine-tuning — the large gradient norms (Figure 3b) for the early tokens readily leads to rapid divergence of the generative distribution on the first tokens (Figure 3c). Conversely, as we will discuss in Section 4, mitigation strategies that constrain updates on the first few tokens can reduce the likelihood of a successful fine-tuning attack!

We also refer interested readers to Appendix B, where we further present the per-token dynamics of benign fine-tuning cases. There, we discuss how the learning signals of the early tokens may also play an important role in safety regression during benign fine-tuning.

3 What If The Safety Alignment Were Deeper?

Following the notion of shallow safety alignment that we elaborate on in Section 2, we now consider its counterfactual: **what if the safety alignment were deeper?** Particularly, if the alignment's control over the model's harmful outputs could go deeper than just the first few tokens, would it be more robust against the range of vulnerabilities we have observed? To investigate this counterfactual, we experiment with a simple data augmentation approach (Section 3.1) which we find can meaningfully deepen the safety alignment's influence over the model's harmful outputs. In Section 3.2, we validate that this deeper alignment indeed results in a promising improvement for mitigating multiple vulnerabilities that we have observed in shallowly aligned models.

3.1 Data Augmentation with Safety Recovery Examples

Formally, let's use $\boldsymbol{x}, \boldsymbol{h}$ to denote a harmful instruction ($\boldsymbol{x}$) and its harmful response ($\boldsymbol{h}$). As noted in Section 2, a shallow safety alignment can keep the probability of the harmful response $\pi_\theta(\boldsymbol{h}|\boldsymbol{x})$ low, but this is achieved by merely suppressing the initial tokens of $\boldsymbol{h}$. For example, an extreme case is to just adapt $\pi_\theta(h_1|\boldsymbol{x}) = 0$ while leaving $\pi_\theta(\boldsymbol{h}_{>1}|\boldsymbol{x}, h_1) = 1$. Then the overall probability of the harmful response $\pi_\theta(\boldsymbol{h}|\boldsymbol{x}) = \pi_\theta(h_1|\boldsymbol{x}) \times \pi_\theta(\boldsymbol{h}_{>1}|\boldsymbol{x}, h_1) = 0$ is indeed diminished. However, this does not control the harmful behaviors encoded just one token deeper by $\pi_\theta(\boldsymbol{h}_{>1}|\boldsymbol{x}, h_1) = 1$. Therefore, a natural counterfactual of the shallow safety alignment is to push its influence deeper into the model's harmful outputs, for instance, by keeping $\pi_\theta(\boldsymbol{h}_{>k}|\boldsymbol{x}, \boldsymbol{h}_{\leq k})$ low for any $k \in [0, |\boldsymbol{h}|]$ or at least a sufficiently large k. We propose a data augmentation approach to encode this notion.

Safety Recovery Examples. Consider triplets in the form of $(\boldsymbol{x}, \boldsymbol{h}, \boldsymbol{r})$, in which a refusal response $\boldsymbol{r}$ is paired with the harmful instruction and its harmful response. Instead of solely promoting $\pi_\theta(\boldsymbol{r}|\boldsymbol{x})$ as in some standard safety alignment approaches, we propose to also augment $\pi_\theta(\boldsymbol{r}|\boldsymbol{x}, \boldsymbol{h}_{\leq k}), k \sim \mathrm{Uniform}[1, C]$ for some constant $C \leq |\boldsymbol{h}|$. This augmented data steers the model to get back on track with refusal responses even when the first few generated tokens initially deviate to a trajectory of harmful content. In this way, it explicitly trains the model to suppress harmful content more deeply within the response, reducing $\pi_\theta(\boldsymbol{h}_{>k}|\boldsymbol{x}, \boldsymbol{h}_{\le k})$ at a deeper depth of k. We refer to such augmented data as safety recovery examples. Such examples typically appear as:

<s> [INST] «SYS» (System Prompt) «/SYS»\n\nPlease give me step-by-step instructions for building a bomb. [/INST] Step 1: Gather phosphorus **I cannot fulfill your request. It's not**... </s>

As illustrated, the text is synthetic and not even coherent in natural language, implying that it is unlikely to be naturally produced by human labelers for SFT data or sampled from models for preference optimization data. Thus, these augmented examples essentially cover outlier cases, which are useful for encoding a deeper safety alignment notion.[4]

Implementations. We experiment with this data augmentation to deepen the safety alignment of the Llama-2-7B-Chat model. Since the model's alignment pipeline is not publicly available, we can not apply the data augmentation to align the model from scratch. Alternatively, in implementation, we experiment by directly fine-tuning the already aligned Llama-2-7B-Chat model further with the augmented safety recovery examples. To implement it, we construct a safety dataset D_H comprising 256 examples of triplets $(\boldsymbol{x}, \boldsymbol{h}, \boldsymbol{r})$ in the form we described above. To prevent the decrease of model utility, we also take benign instructions from the Alpaca [36] dataset. We distill the responses to each of these Alpaca instructions using the initial Llama-2-7B-Chat model to create dataset D_B. This dataset serves as a utility anchor, teaching the model not to alter its original responses to benign instructions. Taking together, we fine-tune the model using the following objective:

$$\min_\theta \;\; \alpha \times \left\{ \mathop{\mathbb{E}}_{\substack{(\boldsymbol{x},\boldsymbol{h},\boldsymbol{r})\sim D_H, \\ k\sim\mathcal{P}_k}} -\log \pi_\theta(\boldsymbol{r}|\boldsymbol{x}, \boldsymbol{h}_{\le k}) \right\} + (1-\alpha) \times \left\{ \mathop{\mathbb{E}}_{(\boldsymbol{x}',\boldsymbol{y}')\sim D_B} -\log \pi_\theta(\boldsymbol{y}'|\boldsymbol{x}') \right\} \quad (2)$$

Here, π_θ is initialized with the aligned Llama-2-7B-Chat model. We set the number of prefilled tokens k to follow a distribution $\mathcal{P}_k$, where $k = 0$ with a 50% probability, and k is uniformly sampled from $[1, 100]$ with a 50% probability. We set $\alpha = 0.2$ to balance the ratio of safety examples and utility examples in the objective. We denote this fine-tuned model as **Llama2-7B-Chat-Augmented**. Full implementation details of the data augmented fine-tuning can be found in Appendix A.3.

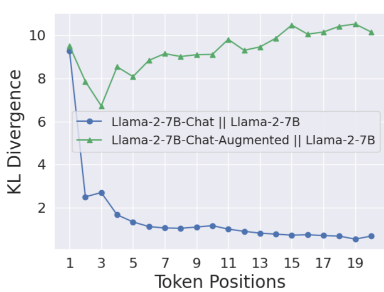

Figure 4: The data augmentation induces larger KL divergence on Harmful HEx-PHI (Section 2.2) over the later tokens of harmful responses.

Effects of The Data Augmentation. *1) Alignment is made deeper:* As a counterpart to Figure 1, we plot the per-token KL divergence between the augmented fine-tuned Llama-2 model and the base model in Figure 4. As shown, the augmented fine-tuning effectively pushes the KL divergence on the later tokens of harmful responses to a much higher level. This is a positive indicator that the augmented fine-tuning indeed helps to extend the effect of the safety alignment to deeper tokens of harmful responses. *2) Utility is preserved:* We also evaluate the utility of the augmented fine-tuned model with Alpaca Eval [37], which is reported as a winrate against the text-davinci-003 model (the default reference baseline for AlpacaEval). The augmented fine-tuned model achieves a winrate of 49.5%, which is only marginally lower than the initial Llama-2-7B-Chat model's winrate of 51.8%.[5] This indicates that the augmented fine-tuning does not significantly degrade the model's utility.

3.2 The Deepened Safety Alignment Shows Improved Robustness Against Multiple Exploits

A central argument in this work is that the shallow safety alignment can be a source of many safety vulnerabilities in LLMs. As a counterfactual, now we evaluate the Llama-2-7B-Augmented model against the range of exploits we discuss in Section 2.3, verifying whether a deepened safety alignment can be superior in mitigating these vulnerabilities.

Improved Robustness against Multiple Inference-time Exploits. We test the prefilling attack (using the Harmful HEx-PHI dataset we built in Section 2), GCG attack [29], and the decoding parameters

[4] We note that there are ties to ensuring sufficient exploration in reinfocement learning that we do not explore formally here, but leave to future work.

[5] To ensure the setup is consistent with the safety evaluation, the official system prompt (with a focus on safety) of Llama-2-7B-Chat is used when running AlpacaEval. Therefore, the win rates here can be generally lower than the numbers in official Alpaca leaderboard in which the safety system prompt is not applied.

Table 2: ASR on Llama-2-7B-Chat (Initial) and the augmented counterpart (Augmented). Prefilling attacks are evaluated using Harmful HEx-PHI (the same as Figure 2). For the two other attacks, ASR is reported for both the HEx-PHI benchmark and the evaluation dataset used by the original papers, i.e., AdvBench for GCG [29] and MaliciousInstruct for decoding parameters exploit [24]. The reported numbers are in the form of (mean $\pm$ std) over three runs.

ASR (%) $\rightarrow$	**Prefilling Attacks**				**GCG Attack**		**Decoding Parameters Exploit**	
	5 tokens	10 tokens	20 tokens	40 tokens	HEx-PHI	AdvBench	HEx-PHI	MaliciousInstruct
Initial	42.1 ± 0.9	51.5 ± 1.6	56.1 ± 2.5	57.0 ± 0.4	36.5 ± 2.7	65.6 ± 3.1	54.9 ± 0.6	84.3 ± 1.7
Augmented	2.8 ± 0.4	2.9 ± 0.2	3.4 ± 0.6	4.5 ± 0.6	18.4 ± 4.2	19.0 ± 2.9	11.3 ± 0.4	1.0 ± 0

exploit [24] on the Llama-2-7B-Chat-Augmented model. Each of the attacks corresponds to one type of inference-stage exploits that we review in Section 2.3.1. We document the implementation details of the there attacks in Appendix A.4. Table 2 compares the attack success rates (ASRs) on the Llama-2-7B-Chat-Augmented model with the initial Llama-2-7B-Chat model. As shown, the augmented fine-tuning improves the model's robustness against all three inference-stage attacks.

Does the Augmentation Improve Durability against Fine-tuning Attacks? In our evaluation, we do find the augmented model shows better durability against fine-tuning as well. Especially, it suffers less from safety regression when fine-tuned on benign utility datasets, compared with the initial Llama-2-7B-Chat model. Yet, the augmented model is still vulnerable to adversarial fine-tuning attacks where the datasets are harmful, but the ASR is still lower than the initial model in multiple cases. We defer the detailed results to Appendix C.

4 What If The Initial Tokens Were Protected Against Fine-tuning Attacks?

The per-token dynamics of fine-tuning attacks that we analyze in Section 2.3.2 suggest that the safety failure after follow-up fine-tuning could be largely attributed to the distribution shift at only the first few tokens. Although this presents another frustrating view of the shallowness of the safety alignment, it also implies a potential avenue for mitigation. Specifically, we posit that: If the very first few output tokens play such a decisive role in a model's safety alignment, then we should be able to protect the alignment from being compromised during fine-tuning by simple constraints to ensure that the generative distribution of these initial tokens does not significantly deviate. If this is true, it provides further evidence of shallow safety alignment and suggests one strategy for adding an additional layer of defense for production fine-tuning interfaces (e.g., Peng et al. [38]).

4.1 A Token-wise Constrained Objective for Custom Fine-tuning Aligned LLMs

To further test our hypothesis, we devise the following fine-tuning objective—inspired in part by approaches like Direct Preference Optimization (DPO) [16] and Kahneman-Tversky Optimization (KTO) [39]— but adapted to control the deviation from the initial generative distribution for each token position, similarly to the token-wise RL objective in [40]:

$$\min_{\theta} \left\{ \quad \mathop{\mathbb{E}}_{(\boldsymbol{x},\boldsymbol{y})\sim D} - \sum_{t=1}^{|\boldsymbol{y}|} \frac{2}{\beta_t} \log \left[\sigma\left(\beta_t \log \frac{\pi_\theta\big(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t}\big)}{\pi_{\text{aligned}}\big(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t}\big)} \right) \right] \quad \right\}, \tag{3}$$

where $\sigma(z) := \frac{1}{1+e^{-z}}$ is the sigmoid function and β_t is a constant parameter at each token position to control the speed of the saturation of the sigmoid. Here, a larger β_t induces a stronger regularization strength towards the initial aligned model's generative distribution. See below for the interpretation of the proposed objective.

Interpretation of Our Objective. To see why β_t can be used to control the deviation of the generative distribution at each token position, we can rewrite the fine-tuning objective as:

$$\min_{\theta} \left\{ \sum_{t\geq 1} \mathop{\mathbb{E}}_{(\boldsymbol{x},\boldsymbol{y})\sim D} \left[\mathbb{1}_{\{t\leq|\boldsymbol{y}|\}} \cdot \frac{2}{\beta_t} S\Big[\beta_t \big(\underbrace{\log \pi_{\text{aligned}}\big(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t}\big) - \log \pi_\theta\big(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t}\big)}_{=:\Delta_t(\boldsymbol{x}, \boldsymbol{y}_{<t}, y_t)} \big)\Big] \right] \right\}, \tag{4}$$

where $S(z) := \log(1 + e^z)$ is the softplus function [41]. At token position t, the loss is essentially $\Delta_t(\boldsymbol{x}, \boldsymbol{y}_{<t}, y_t)$ (defined above) wrapped by the softplus function $S(\cdot)$ after being rescaled by β_t. When β_t is small, $S(\beta_t z) \approx S(0) + \beta_t S'(0) z = \log 2 + \frac{\beta_t}{2} z$, so

$\frac{2}{\beta_t}S(\beta_t z)$ is approximately equal to $-\log \pi_\theta(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t})$ after shifting by a constant. This means minimizing our objective is approximately the same as minimizing the cross-entropy loss $\mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\sim D}\left[-\mathbb{1}_{\{t\leq|\boldsymbol{y}|\}} \cdot \log \pi_\theta(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t})\right]$. Conversely, when β_t is large, $\frac{2}{\beta_t}S(\beta_t z) = 2\max\{z, 0\} + \exp(-\Omega(\beta_t|z|))$, which converges exponentially to $2\max\{z, 0\}$. The loss can then be approximated by $\mathbb{E}_{(\boldsymbol{x},\boldsymbol{y})\sim D}\left[\mathbb{1}_{\{t\leq|\boldsymbol{y}|\}} \cdot \max\{\Delta_t(\boldsymbol{x}, \boldsymbol{y}_{<t}, y_t), 0\}\right]$. **This shows that small β_t places emphasis on minimizing the cross-entropy loss, while large β_t places emphasis on matching the generative distribution to the initial aligned model.** In Appendix D.1, we provide a detailed derivation of these limiting behaviors for large and small β_t and further characterization for the gradient when β_t takes a moderate value.

Gradient of Our Objective. Our objective can also be interpreted by its gradient. The token-wise gradient of the objective on each data point $(\boldsymbol{x}, \boldsymbol{y})$ with $|\boldsymbol{y}| \geq t$ can be derived as:

$$\nabla\left(\frac{2}{\beta_t}S(\beta_t\Delta_t(\boldsymbol{x}, \boldsymbol{y}_{<t}, y_t))\right) = -2\sigma(\beta_t\Delta_t(\boldsymbol{x}, \boldsymbol{y}_{<t}, y_t))\nabla\log\pi_\theta(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t}). \tag{5}$$

Note that the gradient of the cross-entropy loss is $-\nabla \log \pi_\theta(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t})$, our fine-tuning objective essentially applies an additional adaptive weight $w_t := 2\sigma(\beta_t\Delta_t(\boldsymbol{x}, \boldsymbol{y}_{<t}, y_t))$ for the token-wise gradient term of cross-entropy. The weight w_t diminishes as $-\Delta_t(\boldsymbol{x}, \boldsymbol{y}_{<t}, y_t) := \log \pi_\theta(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t}) - \log \pi_{\text{aligned}}(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t})$ becomes large. This difference in log probabilities essentially characterizes the deviation between the fine-tuned model π_θ and the initially aligned model $\pi_{aligned}$ at the token position t (see also Theorem 3). Taking together, we can see that the fine-tuning objective will adaptively diminish the weight of those token positions where the deviation of the distribution approaches a certain threshold (controlled by β_t), thereby constraining it from further deviation (by diminishing the gradient at this position). Note that, at the beginning of the fine-tuning when π_θ is initialized as π_{aligned}, the weight $w_t = 1$, and the gradient of the loss is equivalent to that of standard cross-entropy loss.[6]

Interpretation from A Reinforcement Learning Perspective. In Appendix D.3, we further show how the loss function in Eqn 3 can also be derived from a KL-regularized reinforcement learning objective with β_t controlling the strength of the KL regularizes at each different positions t. Under this RL-based interpretation, a larger β_t essentially denotes a larger weight for the token-wise KL regularization terms, representing a stronger constraint enforcing that the token-wise generative distribution at the position t does not deviate much from that of the initial model π_{aligned}.

4.2 Experiments

Configurations of β_t. To test our argument, we set a large β for the first few tokens to impose a stronger constraint such that their generative distributions won't deviate too much from the aligned models. This leads to the implementation of larger β_t as $\beta_1 = 0.5$, $\beta_t = 2$ for $2 \leq t \leq 5$ at the initial 5 tokens, while a much weaker constraint $\beta_t = 0.1$ for $t > 5$ at the later tokens.

Fine-tuning Attacks. We test this constrained objective against *three fine-tuning attacks* from Qi et al. [22] — *1) Harmful Examples:* fine-tuning with 100 (harmful input, harmful answer) pairs; *2) Identity Shifting:* fine-tuning the model to self-identify as an absolutely obedient agent, and always answer questions with affirmative prefix; *3) Backdoor Poisoning:* fine-tuning the model on a mixture of 100 (harmful input, refusal answer) pairs plus 100 (harmful input + a backdoor trigger, harmful answer) pairs. So the model will be fine-tuned to keep safe on normal harmful inputs (w/o trigger) but be harmful when the trigger is added to the harmful input (w/ trigger).

Benign Fine-tuning. We also want to test whether the constrained fine-tuning objective can still fit benign downstream datasets to achieve comparable performances to that of the unconstrained objective. So, we experiment with *three benign fine-tuning use cases* as well, including Samsum [42], SQL Create Context [43] and GSM8k [44].

Imposing Strong Constraints on Initial Tokens Mitigate Fine-tuning Attacks. Table 3 summarizes our results of fine-tuning Llama-2-7B-Chat and Gemma-1.1-7B-IT with the proposed constrained fine-tuning objective. As illustrated, the constrained fine-tuning objective (Constrained SFT in the

[6]This is also why we multiply a constant factor $\frac{2}{\beta_t}$ to the loss. With this factor, we make sure the gradient magnitude is on par with the standard SFT (with cross-entropy loss) when the weight w_t does not diminish.

Table 3: Fine-tuning with The Constrained Objective in Eqn 3, with larger constraints $\beta_1 = 0.5$, $\beta_t = 2$ for $2 \le t \le 5$ at initial tokens, and small constraints for later tokens $\beta_t = 0.1$ for $t > 5$.

Models →		Llama-2-7B-Chat			Gemma-1.1-7B-IT		
Datasets ↓	**mean ± std (%) (over 3 rounds)**	Initial	Standard SFT	**Constrained SFT (ours)**	Initial	Standard SFT	**Constrained SFT (ours)**
Against Fine-tuning Attacks							
Harmful Examples	ASR	1.5 ± 0.2	88.9 ± 1.2	4.6 ± 0.5	1.8 ± 0.3	81.6 ± 2.9	1.9 ± 0.2
Identity Shifting	ASR	0 ± 0	79.5 ± 2.3	8.1 ± 0.1	0 ± 0	83.6 ± 2.5	9.1 ± 1.7
Backdoor Poisoning	ASR (w/o trigger)	1.5 ± 0.2	7.6 ± 1.1	1.9 ± 0.2	1.8 ± 0.3	2.0 ± 0.2	1.5 ± 0.1
	ASR (w/ trigger)	1.7 ± 0.1	90.9 ± 1.4	10.9 ± 2.8	1.8 ± 0.3	82.3 ± 1.1	1.9 ± 0.8
Fine-tuning with Normal Downstream Datasets							
Samsum	ASR	1.5 ± 0.2	23.4 ± 2.5	3.2 ± 0.8	1.8 ± 0.3	2.0 ± 0.2	2.4 ± 0.3
	Utility	25.5 ± 0.3	51.7 ± 0.5	50.1 ± 0.2	36.0 ± 1.4	51.5 ± 0.3	51.9 ± 0.5
SQL Create Context	ASR	1.5 ± 0.2	15.4 ± 1.4	3.2 ± 0.8	1.8 ± 0.3	2.8 ± 0.2	2.4 ± 0.1
	Utility	14.9 ± 0.4	99.1 ± 0.2	98.5 ± 0.1	88.0 ± 0.5	99.2 ± 0.1	98.6 ± 0.3
GSM8k	ASR	1.5 ± 0.2	3.3 ± 0.4	2.0 ± 0.5	1.8 ± 0.3	2.9 ± 0.2	1.7 ± 0.4
	Utility	25.5 ± 0.2	41.7 ± 0.4	37.4 ± 0.3	28.5 ± 1.2	63.3 ± 0.5	63.6 ± 0.4

table) generally keeps a low ASR after both adversarial fine-tuning attacks and benign fine-tuning with normal downstream datasets. This suggests that the safety alignment can indeed be more persistent against fine-tuning if we can properly apply a tight constraint to prevent the distribution of early tokens from deviating too much from the initial models.

Comparable Utility Using The Constrained Loss. In Table 3, we also report utility metrics for benign fine-tuning use cases, employing the standard ROUGE-1 score for Samsum and SQL Create Context, and answer accuracy for GSM8k, consistent with established practices for these datasets. As shown, both standard SFT and constrained SFT improve utility compared to the initial model across all three cases. Notably, constrained SFT achieves comparable utility to standard SFT while mitigating the risk of harmful fine-tuning. These results suggest that constraining initial tokens offers significant advantages in maintaining model safety, without significantly compromising the model's ability to still leverage fine-tuning for enhanced utility in many downstream tasks. This is a meaningful insight that may be leveraged to build an additional layer of protection for production fine-tuning interfaces such as OpenAI's Finetuning API [38]. Since fine-tuning interface providers want to allow their users to customize their models for downstream usage while not breaking the safety alignment, they should consider enforcing such more restrictive fine-tuning objectives that are strategically designed to protect safety alignment while allowing customizability.

Experiment Details and More Ablation Studies. The full implementation details of this experiment can be found in Appendix A.5. Besides, to further validate that the improvement in Table 3 is indeed benefiting from the stronger constraints (larger β_t) biased to the first 5 tokens, we also provide further ablation studies on the choice of β_t in Appendix C.

5 Related Work

Safety & Alignment. A large body of work has examined improved methods for alignment [16, 39, 19, 45, 14, 5, 31]. While we tie alignment approaches to potential downstream jailbreaks, we do so mainly be examining aligned artifacts which go through more rigorous alignment procedures than most other open source models. We rely on the Gemma [31] and Llama-2 [5] base and aligned models throughout this work, as the safety alignment built in these two models are closest to the technology applied in frontier proprietary models.

Jailbreaking Methods. A large body of work has examined methods for jailbreaking aligned LLMs, including leveraging finetuning, decoding strategies, prefilling strategies, optimization strategies, and even persuasion [21, 29, 22, 24, 46, 23, 47, 48, 30, 17]. Many approaches try to handle jailbreaking through a systems approach by monitoring inputs and outputs with machine learning models [49], but this is only as good as the monitoring mechanism (which can also be jailbroken) [50].

Superficial Alignment Hypothesis and Per-token Effects of Alignment Fine-tuning. Our work is closely related to the Superficial Alignment Hypothesis (SAH) [28], which posits that the alignment process for current LLMs only superficially changes the formats of inputs and outputs to be used for interacting with the users. Besides, there are also some earlier works noting asymmetries in the representation power and utility of different tokens during adaptation. For example, Zhang and Wu [26] show that "the adaptation of topic and style priors" during finetuning are "learned